

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO3: *Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

Assessment Tool Weightage: 50%

Minutes

QUESTIONS: 30

Time Duration: 30

Total Marks:30

Annexure A

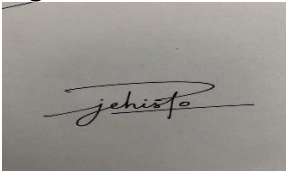
MCQ

Code of Conduct:

I Jehisto rijin J/ 192511022 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Annexure A

Q1. Which of the following best describes a Logical Agent in AI?

- a) An agent that reacts randomly to the environment c) An agent that only follows pre-defined scripts
- b) An agent that uses a knowledge base and logical inference to decide actions d) An agent that learns from rewards

Q2. In Propositional Logic, ' $P \rightarrow Q$ ' is logically equivalent to:

- a) $P \wedge \neg Q$ c) $\neg Q \rightarrow \neg P$ only
- b) $\neg P \vee Q$ d) Both b and c

Q3. Conjunctive Normal Form (CNF) requires the formula to be expressed as: a)

- A disjunction of conjunctions c) A conjunction of literals only
- b) A conjunction of disjunctions (clauses) d) A sequence of implications

Q4. In FOL, the sentence $\forall x: \text{Student}(x) \rightarrow \text{Passes}(x)$ means:

- a) Some students pass c) If something passes, it is a student
- b) Every entity that is a student passes d) There exists at least one student who passes

Q5. Unification in FOL is the process of finding substitutions that:

- a) Convert clauses to CNF
- b) Make two logical expressions syntactically identical
- c) Prove a goal using backward chaining
- d) Derive facts using Modus Ponens

6. What is the core function of a logical agent in AI?

- a) Reacting only to current percepts
- b) Using logic and knowledge representation to reason and make decisions
- c) Performing physical actions without internal representation
- d) Relying solely on sensors for navigation

7. Which component of a logical agent is responsible for storing facts and rules about the world? a)

- Inference Engine
- b) Perception Unit
- c) Knowledge Base (KB)
- d) Action Selector

8. What does a knowledge-based agent do immediately after obtaining a percept from the environment?

- a) Selects an action
- b) Executes the action
- c) Updates the Knowledge Base with the new percept
- d) Reasons to derive new information

9. Which operation is used to add new information to a Knowledge Base?

- a) ASK b) TELL c) PERCEIVE d) UPDATE

10. What is a proposition in formal logic?

- a) A command or request
- b) A declarative statement that is either true or false
- c) A question regarding the environment
- d) A letter representing a variable

11. Which of the following is NOT a proposition?

- a) "The sun rises in the east."
- b) "5 + 3 = 10."
- c) "Close the door."
- d) "Chennai is in India."

12. Which logical operator is represented by the symbol $\neg$?

- a) AND
- b) OR
- c) NOT
- d) Implication

13. The AND ($\wedge$) operator is true only when:

- a) At least one proposition is true
- b) Both propositions are true
- c) Both propositions have different truth values
- d) Neither proposition is true

14. A proposition that is always true, regardless of the truth values of its variables, is called a:

- a) Contradiction
- b) Tautology
- c) Contingency
- d) Equivalence

15. In an implication ($P \rightarrow Q$), the statement is false only when:

- a) P is false and Q is true
- b) P is true and Q is false
- c) Both P and Q are false
- d) Both P and Q are true

16. What is the number of rows in a truth table for a logical expression with 3 propositions? a) 4 b) 6 c) 8 d) 9

17. Which of these represents one of De Morgan's Laws?

- a) $\neg(P \wedge Q) \equiv \neg P \vee \neg Q$
- b) $\neg(P \wedge Q) \equiv \neg P \wedge \neg Q$
- c) $\neg(\neg P) \equiv P$
- d) $P \vee \neg P \equiv \text{True}$

18. A proposition that is sometimes true and sometimes false is known as a:

- a) Tautology
- b) Contradiction
- c) Contingency
- d) Implication

19. Which logical connective is true if and only if both propositions have the same truth value?

- a) Implication ($\rightarrow$) b) Biconditional ($\leftrightarrow$) c) OR ($\vee$) d) NOT ($\neg$)

20. According to operator precedence, which operator should be evaluated first?

- a) AND ($\wedge$) b) OR ($\vee$) c) NOT ($\neg$) d) Implication ($\rightarrow$)

21. What is the process of deriving a conclusion from given premises called?

- a) Perception b) Representation c) Inference d) Predication

22. Which inference rule follows the form: If P, then Q; P; Therefore, Q?

- a) Modus Tollens b) Modus Ponens c) Hypothetical Syllogism d) Resolution

23. Which valid reasoning pattern is represented by: $P \rightarrow Q$; $\neg Q$; Therefore, $\neg P$?

- a) Modus Ponens b) Disjunctive Syllogism c) Modus Tollens d) Addition

24. What can be concluded from the premises $(P \rightarrow Q)$ and $(Q \rightarrow R)$ using Hypothetical Syllogism? a)

- $P \wedge R$ b) $P \vee R$ c) $P \rightarrow R$ d) $\neg P \rightarrow \neg R$

25. The rule that states if "P or Q" is true and "Not P" is true, then "Q" must be true is called:

- a) Simplification b) Conjunction c) Disjunctive Syllogism d) Resolution

26. Which rule allows you to conclude "P" from the premise "P and Q"?

- a) Addition b) Simplification c) Conjunction d) Modus Ponens A

27. The inference rule Resolution follows which symbolic form?

- a) $P \therefore P \vee Q$ b) $P, Q \therefore P \wedge Q$ c) $P \vee Q, \neg P \vee R \therefore Q \vee R$ d) $P \rightarrow Q, Q \rightarrow R \therefore P \rightarrow R$

28. Which of the following is an example of the "Addition" rule?

- a) P and Q, therefore P
b) P, therefore P or Q
c) P, Q, therefore P and Q
d) P or Q, not P, therefore Q

29. "If it rains, the ground is wet. The ground is wet. Therefore, it rained." This is an example of which invalid reasoning pattern?

- a) Denying the Antecedent b) Affirming the Consequent c) Modus Ponens d) Double Negation

30. What is the name of the invalid pattern: $P \rightarrow Q$; $\neg P$; Therefore, $\neg Q$?

- a) Affirming the Consequent b) Denying the Antecedent c) Modus Tollens d) Hypothetical Syllogism

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	30%	Demonstrates thorough understanding; answers all questions accurately	Shows good understanding with few errors	Partial understanding; several incorrect responses	Lacks understanding; majority of answers incorrect
Accuracy of Answers	25%	All answers are correct	Most answers are correct with minor mistakes	Some correct answers; frequent errors	Mostly incorrect answers
Application of Knowledge	20%	Correctly applies concepts to problem-based questions	Applies concepts with minor errors	Limited ability to apply concepts	Unable to apply concepts
Time Management	15%	Completes quiz well within time efficiently	Completes on time with minor delays	Requires extra time or rushes through questions	Unable to complete within given time
Question Interpretation	10%	Interprets all questions correctly and responds appropriately	Misinterprets a few questions	Often misinterprets questions	Unable to understand questions

CO3 - AT2 - MCQ test

Name: Pavan P

Reg no: 192411017

1. b) ✓
2. b) ✓
3. b) ✓
4. b) ✓
5. b) ✓
6. b) ✓
7. c) ✓
8. c) ✓
9. b) ✓
10. b) ✓
11. c) ✓
12. c) ✓
13. b) ✓
14. b) ✓
15. b) ✓
16. c) ✓
17. a) ✓
18. c) ✓
19. b) ✓
20. c) ✓
21. c) ✓
22. b) ✓
23. c) ✓
24. c) ✓
25. c) ✓
26. b) ✓
27. c) ✓
28. b) ✓
29. b) ✓

30) b) ✓

29
30

2

